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The Qang (qg): A Unified Angular-Probability Unit and Metric for Parametric Quantum Circuit Design

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Abstract

In contemporary quantum information processing, the parameterization of single-qubit states relies on polar coordinates on the Bloch sphere, while measurement outcomes are expressed as statistical expectation values or collapse probabilities. We propose the qang (symbol: qg), a normalized dimensionless unit designed to bridge the geometric rotation angle of a state vector directly with its measurement probability distribution. We present two formal operational definitions: the **Polar Bias Qang** (qg_Z), anchored to the expectation value of the Pauli-Z observable, and the **Superposition Dispersion Qang** (qg_S), mapping polar deviation to the binary Shannon entropy of measurement outcomes. We demonstrate its utility in gradient-based optimization algorithms (VQE, QAOA) and discuss how defining a domain-specific unit streamlines compilation, noise characterization, and hardware benchmarking.

1. Introduction and Motivation

In quantum computing and quantum state engineering, parameterized single-qubit rotations serve as the fundamental building blocks of variational circuits (Ansätze), quantum machine learning models, and fault-tolerant compilation pipelines.

Under the standard Bloch sphere representation, an arbitrary pure single-qubit state is parameterized as:

$$|\psi\rangle = \cos(\theta / 2)|0\rangle + e^{i\varphi} \sin(\theta / 2)|1\rangle$$

where $\theta \in [0, \pi]$ represents the polar angle and $\varphi \in [0, 2\pi)$ is the azimuthal phase angle.

When executing algorithms, researchers repeatedly oscillate between two distinct mathematical representations:

1. Continuous physical control parameters (microwave pulse duration, phase shifts, rotation angles in radians or degrees).
2. Probabilistic projection profiles obtained through Born's rule upon measurement:
 - $P(|0\rangle) = |\langle 0|\psi\rangle|^2 = \cos^2(\theta / 2)$
 - $P(|1\rangle) = |\langle 1|\psi\rangle|^2 = \sin^2(\theta / 2)$

Currently, bridging these domains requires explicit trigonometric conversions. To provide an intuitive, standardized metric that encapsulates both geometric displacement and probabilistic outcome bias, we introduce the **qang** (abbreviated as **qg**).

2. Mathematical Formalization

We formulate the qang under two complementary definitions:

2.1 Polar Bias Qang (qg_Z)

The primary formulation of the qang quantifies the measurement bias between computational basis states $|0\rangle$ and $|1\rangle$ as a function of the polar angle θ :

$$\text{qg_Z}(\theta) = P(|0\rangle) - P(|1\rangle) = \cos^2(\theta / 2) - \sin^2(\theta / 2) = \cos(\theta)$$

Equivalently, in terms of quantum mechanics operators, qg_Z corresponds to the expectation value of the Pauli-Z operator:

$$\text{qg_Z} = \langle \psi | \sigma_z | \psi \rangle$$

Inversion Relation

Given a target polarization or expectation value $\text{qg_Z} \in [-1, +1]$, the required polar rotation angle θ is directly recovered via:

$$\theta = \arccos(\text{qg_Z})$$

Standard Anchor Points

State Description	Polar Angle (θ)	Qang Value (qg_Z)
North Pole ($ 0\rangle$)	0 rad	+1.0 qg
Equator / Maximal Superposition ($ +\rangle, -\rangle$)	$\pi/2$ rad	0.0 qg
South Pole ($ 1\rangle$)	π rad	-1.0 qg
Canonical T-gate state	$\pi/4$ rad	$\approx +0.7071$ qg

2.2 Measurement-Outcome Entropy Qang (qg_S)

For tasks quantifying quantum coherence loss or state purity relative to maximal superposition, the entropic formulation normalizes the state's polar projection against the binary Shannon entropy of the Z-basis measurement outcome distribution. Note that this is the classical Shannon entropy of the Born-rule outcome probabilities, not the von Neumann entropy of the quantum state itself — for any pure state $|\psi\rangle$, $S(\rho) = -\text{Tr}(\rho \log \rho)$ is identically zero regardless of θ , since $\rho = |\psi\rangle\langle\psi|$ has eigenvalues $\{1, 0\}$. The quantity below is the entropy of the classical distribution obtained after measurement:

$$qg_S(\theta) = -P(|0\rangle) \log_2(P(|0\rangle)) - P(|1\rangle) \log_2(P(|1\rangle))$$

In this scale:

- Deterministic classical poles ($|0\rangle, |1\rangle$) correspond to $qg_S = 0.0$ qg.
- Maximum quantum uncertainty ($\theta = \pi/2$) corresponds to $qg_S = 1.0$ qg.

Domain and Invertibility: Because the binary entropy function is symmetric, $H(p) = H(1-p)$, $qg_S(\theta) = qg_S(\pi-\theta)$: a single qg_S value does not uniquely determine θ over the full domain $[0, \pi]$. A well-defined inversion $\theta = H^{-1}(qg_S)$ therefore requires restricting to a half-domain, e.g. $\theta \in [0, \pi/2]$.

State Description	Polar Angle (θ)	Qang Value (qg_S)
North Pole ($ 0\rangle$)	0 rad	0.0 qg

State Description	Polar Angle (θ)	Qang Value (qg_S)
Equator / Maximal Superposition ($ +\rangle, -\rangle$)	$\pi/2$ rad	1.0 qg
South Pole ($ 1\rangle$)	π rad	0.0 qg
Canonical T-gate state	$\pi/4$ rad	≈ 0.601 qg

Table 2. Anchor values of the entropic qang (qg_S). Note that both poles map to the same value (0.0 qg), reflecting the non-invertibility discussed above.

2.3 Relation to Existing Formalisms

It is important to state plainly what is and is not new here. $\text{qg_Z}(\theta) = \cos(\theta)$ is numerically identical to $\langle \sigma_z \rangle$, the expectation value of the Pauli-Z operator — equivalently, the z-component of the Bloch vector — which is already a standard quantity throughout quantum information theory (Nielsen & Chuang, 2010). Likewise, $\text{qg_S}(\theta)$ is the ordinary binary Shannon entropy $H(p)$ applied to the Born-rule outcome distribution. Neither qg_Z nor qg_S is a new physical quantity.

The contribution proposed in this work is instead a named, normalized engineering unit built on top of these existing quantities — fixed anchor points, an explicit inversion relation, and a millesimal subunit (the milliqang) intended for direct use in circuit design, calibration and optimization tooling. This is analogous to how the decibel (dB) is not a new physical quantity but a scaled, named convention over a pre-existing power ratio, or how the musical "cent" is a named convention over a logarithmic frequency ratio. The qang's value is proposed to lie in this same category: a practical, standardized unit of convenience rather than a physical discovery.

3. Key Properties and Advantages

- 1. Direct Operational Meaning:** Expressing changes in qg immediately informs the circuit designer of the resultant probability shift without requiring mental or algorithmic trigonometric transformation.

- 2. **Phase Decoupling:** The polar qang cleanly isolates population probability dynamics from phase shifts ϕ , simplifying the tuning of non-commuting operations.
- 3. **Scale Invariance:** The metric is normalized to $[-1, +1]$, aligning naturally with hardware cost functions, loss landscapes, and neural network activation profiles.

4. Application Domains

4.1 Variational Quantum Eigensolvers (VQE) & QAOA

In parameterized quantum circuits, optimization trajectories often suffer from barren plateaus or ill-conditioned gradients when parameterized purely in radians. Formulating gradients with respect to qang space:

$$\partial E / \partial(\text{qg}) = (\partial E / \partial \theta) * (d\theta / d(\text{qg})) = -(1 / \sin(\theta)) * (\partial E / \partial \theta)$$

provides an alternative parameter landscape that highlights population transfer rates.

Limitation: This reparametrization is not free of cost: since $d(\text{qg_Z})/d\theta = -\sin(\theta)$, the Jacobian vanishes exactly at the poles $\theta = 0, \pi$ — precisely the ± 1.0 qg anchor points in Table 1 — so $\partial E/\partial(\text{qg})$ diverges there. The qang reparametrization trades the known difficulties of θ -space (e.g. barren plateaus) for a coordinate singularity of its own near the poles, and should be used away from that region, or combined with an explicit regularization near $|\text{qg_Z}| \rightarrow 1$.

4.2 Quantum Error Characterization & Calibration

Drift in microwave pulse amplitudes leads to over-rotation or under-rotation errors (e.g., coherent systematic errors). Expressing drift in milliqangs (m-qg) establishes a direct error tolerance threshold on observable fidelity.

Limitation: This tolerance is not uniform across the Bloch sphere: because $|d(\text{qg_Z})/d\theta| = \sin(\theta)$, the qang scale is most sensitive to angular drift at the equator ($\theta = \pi/2$) and least sensitive near the poles, where it approaches zero. A fixed milliqang tolerance therefore corresponds to a much looser angular tolerance near $|0\rangle$ and $|1\rangle$ than near a superposition state, and calibration budgets expressed in milliqangs should state the operating point θ at which they apply.

5. Summary

The qang establishes a pragmatic bridge between geometric qubit parameterization and experimental readout, expressed as a named, normalized unit built on already-established quantities ($\langle\sigma_z\rangle$ and binary Shannon entropy) rather than as a new physical observable.

6. Future Research Directions

The formalization above leaves open several concrete lines of work, organized around the four areas identified as most immediately actionable for the qang framework:

- Quantum-code treatment of the qang as a unit: implement qg as a native unit/type in quantum-computing SDKs (Qiskit, Cirq, PennyLane wrappers), so that circuit parameters, measurement results, and optimizer state can be constructed, converted, and displayed natively in qg / m-qg rather than requiring manual trigonometric conversion at each call site.
- Formal and empirical validation of the qang: prove the stated bounds, the domain over which qg_Z and qg_S are each invertible (Section 2.2), and error-propagation bounds when converting between θ -space and qg-space; complement this with empirical benchmarking of qang-parameterized optimizers on NISQ hardware and simulators.
- Characterization of qang gradients: study and, where possible, regularize the coordinate singularity at $\theta = 0, \pi$ identified in Section 4.1, and compare the convergence behavior of qg-space optimization against standard θ -space parameterization and quantum natural gradient methods.
- Generalization of qang probabilities: extend the binary framework to POVMs and multi-outcome measurements, to mixed states (where, unlike the pure-state case treated here, the von Neumann entropy of ρ is no longer trivially zero), and to multi-qubit systems via tensor-product projection profiles.

Code Availability

A reference implementation of the qang unit (qg_Z, qg_S, their inversion relations, and the milli-qang subunit) is available at <https://github.com/Viny2030/quang>, together with a test suite that reproduces the anchor values in Tables 1 and 2 and the gradient singularity discussed in Section 4.1.

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